RITIK KUMAR

5th Year, BS+MS (Research) Physics
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EDUCATION

Indian Institute of Science, Bengaluru, India. Aug. 2022 – July 2027 (expected)
(BS+MS) in Physics

RESEARCH EXPERIENCE

Tracing Outflows Across Different Gas Phases in AGN Sep. 2025 – Apr. 2026
Bachelor's Thesis
*Advisor: **Prof. C. S. Stalin**, IIA, Bengaluru*

- Systematically investigating ionized outflows in AGN through X-ray analysis of sources from the Milli Quasar Catalog cross-correlated with XMM-Newton archives. I cross-correlated the milli-quasar catalog with XMM-Newton at 5".
- Combining X-ray detections to explore connections between different outflow phases and operational scales in AGN
- We identified several AGN for which detailed outflow studies are not yet available; currently performing spectral modelling of their XMM-Newton data and we discovered several systems with multiple outflow components at 0.06–0.33c—rare "UFO forests" challenging simple wind models and we are still conducting full analysis.
- Modeling methodology: Began with a simple power-law continuum, then incorporated reflection components to isolate residual features associated with outflows. After identifying line-like signatures, Gaussian absorption/emission models (gabs, zgauss) were applied for detection. Finally, physical modeling was performed using zxipcf and XSTAR grids for the final spectral analysis.
- We have found the sources which are showing the warm absorption and UFO from Fe K line. And interesting thing is, we have found 2 to 3 source for the first time and these sources have soft x-ray ultrafast outflows like the paper by J N Reeves found in PDS 456. To find these signature and model them is not possible with avaliable models like zxipcf, therefore, I developed the XSTAR grid to model these signature.
- **Manuscript is in preparation based on the source** "*****" where we have found soft x-ray ultra-fast outflows and Fe K line outflows together using XSTAR (photoionised grid) and this is case of **multiphase outflows.**: soft x-ray UFO velocity 0.22c and Fe K line UFO velocity is 0.32c.

Spectral Analysis of Seyfert Galaxies Dec. 2024 – Jan. 2026
*Advisor: **Prof. C. S. Stalin**, IIA, Bengaluru*

This is my independent work with Prof C. S. Stalin.

- Developed a Python pipeline to cross-match NLSy1/BLSy1 catalogs with NuSTAR observations (12 radius) using angular-separation and signal-to-noise criteria.
- Reduced NuSTAR data with NuSTARDAS (FPMA/FPMB), including calibration, background filtering, and barycentric corrections.
- Performed broadband spectral modeling in XSPEC using cutoffpl, nthcomp, pexrav, xillverCp, and relxill to constrain coronal temperature, high-energy cutoff, and reflection properties.
- Measured coronal temperatures across multiple epochs with well-constrained uncertainties, finding evidence for temporal stability in several sources.
- **Reported coronal temperature measurements for multiple sources for the first time. And setup the geometry, thickness of corona, reflection fraction, variability and one of the important source is ****** where we have seen the coronal temperature is varying over epochs means disk is not stable.**
- ***Manuscript in preparation.***

Lead Researcher – Relativistic Reflection Modeling in AGN Jan. 2026 – Present (Ongoing)
Full Member, ECAP X-ray Research Group
Collaboration with **Prof. Jörn Wilms, Dr. Thomas Dauser,** , Alexey Nekrasov
*Erlangen Centre for Astroparticle Physics **(ECAP)**, Friedrich-Alexander-Universität, Germany*

  - Leading an independent research project on developing and testing new relativistic reflection (relxill_ring) models for AGN X-ray spectroscopy.
  - **Granted full membership in the ECAP X-ray research group (March 2026)**, including:
    * Full SSH access to Remeis Observatory computing cluster for large-scale data analysis
    * Active membership in Sternwarte X-ray research mailing list
    * Regular participation in weekly ECAP X-ray group research meetings
  - Serving as lead researcher: Got the source, performed complete data reduction pipelines (calibration, filtering, spectral extraction) for XMM-Newton and NuSTAR, and conducting broadband spectral modeling under guidance from ECAP group.
  - Testing and validating new relativistic reflection model variants (extended corona, ring-like corona given by Nekrasov, A. D, et al. 2025), with real observational data, analyzing reflection signatures, disk-corona geometry, height of corona, and spin measurements through detailed XSPEC modeling

Light Curve Reconstruction of Gamma-Ray Bursts — Apr. 2025 – Present (Ongoing)
*Advisor: **Prof. Maria G. Dainotti**, NAOJ, Japan*

Collaborative group project. My contributions:

- Developed and optimized Deep Gaussian Process pipelines for reconstruction of gamma-ray burst (GRB) light curves.
- Applied Deep Gaussian Process models to 545 GRBs, including highly sparse and noisy datasets with limited observational coverage.
- Developed methods for uncertainty estimation and robust inference in irregular astrophysical time-series data while minimizing false detections.
- Contributed to computational workflow optimization, model comparison, and statistical analysis across multiple machine learning reconstruction frameworks.
- Kaushal, Aditi, M.G. Dainotti, **Ritik Kumar, et al.** "Multi-Model Framework for Reconstructing Gamma-Ray Burst Light Curves," *Journal of High Energy Astrophysics (JHEAp)*
  DOI: 10.1016/j.jheap.2025.100519

- Currently, we are working on our next paper of this series using different ML models.
- Research conducted under the mentorship of Prof. Maria G. Dainotti in an international collaboration; **testimonial available upon request**.

Physics-Informed Neural Networks for ODEs/PDEs — Jan. 2025 – Apr. 2025
*Advisor: **Prof. Rishita Das**, IISc, Bengaluru*

- Implemented PINNs in PyTorch to capture discontinuities in shock tubes.
- Compared PINNs with traditional numerical methods (Runge-Kutta, Euler Method, central difference), which proved more reliable for handling discontinuities.
- Solved 1D/2D/3D Burgers' equations using different initial and boundary conditions, Poisson equation. We matched our results of PINNs with Numerical method and they were nearly the same.
- Solved the 1D Euler's equation using PINNs with initial Sod Shock Tube, and we captured partial shocks.
- Applied CFL stability criterion (CFL = 0.5) with 400 grid points for stable time-stepping in compressible flow simulations with shock wave propagation.

PUBLICATIONS & PRESENTATIONS

Journal Articles

- Kaushal, Aditi,M.G. Dainotti, **Ritik Kumar**, et al. (2025). *Multi-Model Framework for Reconstructing Gamma-Ray Burst Light Curves*. **Journal of High Energy Astrophysics (JHEAp)**.
  DOI: 10.1016/j.jheap.2025.100519

Manuscripts in Preparation

- **Ritik Kumar**, C. S. Stalin, and J. Hota, "Variation in the Temperature of the Corona in Seyfert-type AGN And disk geometry," 2026.

- **Ritik Kumar**, C. S. Stalin, and J. N. Reeves, "A possible multi-phase outflows in ****.

- **Ritik Kumar**, Ayush, M.G. Dainotti, et al. "Light Curve Reconstruction of Gamma-Ray Bursts"

Conference Presentations

– **XRISM International Conference 2025, Japan** — Poster Presentation
A comparative analysis of the coronal properties of Broad Line and Narrow Line Seyfert 1 galaxies.

– **Astronomical Society of India, Conference 2026, Guwahati** — Poster Presentation
Nature of Coronae in Bright Active Galactic Nuclei and affecting the surrounding medium.

RELEVANT COURSES

Joint Astronomy Programme (JAP) – PhD-level collaborative program (IISc, ISRO, IIA, RRI)

- PH 217: Fundamentals of Astrophysics
- PH 362: Radiative Process in Astrophysics
- PH 363: Introduction to Fluid Mechanics and Plasma Physics
- PH 354: Computational physics
- PH 365: Galaxies and Interstellar Medium
- PH 371: General Relativity and Cosmology
- PH 377: Astronomical Techniques (Seminar Course)
- Machine learning

WORKSHOPS

Fast Radio Bursts (FRBs) Workshop Oct. 6–17, 2025
International Centre for Theoretical Sciences (ICTS), India
Advanced workshop on FTSky: transient signal detection, radio data analysis, and multi-wavelength time-domain astrophysics.

Galaxies and Related Topics: Honoring Prof. Biman Nath Jun. 6–7, 2025
RRI, India

TECHNICAL SKILLS

X-ray Analysis: XSPEC, ISIS, HEASoft, NuSTARDAS, SAS; broadband spectral modelling and timing analysis.

Programming: Python (NumPy, SciPy, Astropy, Matplotlib, Pandas).

Data Pipelines: Python-based automation for NuSTAR/XMM reduction, light curve extraction, spectral extraction, multi-epoch AGN analysis.

Statistical/Machine Learning: Deep Gaussian Processes, Physics-Informed Neural Networks (PINNs), ReFANN, Numerical methods, uncertainty modelling,etc. Time-series inference for irregular and noisy data; model comparison under stochastic variability, MCMC.

Additional Tools: dsplot, DS9, fv, topcat, Astroquery, SDSS (optical) data access.

LEADERSHIP AND SERVICE

- **Management Coordinator**, Pravega (India's largest science and cultural festival) 2024 – 2025
  Indian Institute of Science, Bengaluru.
- **Member**, IISc Astrophysics Club "Astrae" arXiv Club of IISc.
- **Member**, ECAP X-ray Research Group, Germany.
- **Member**, AstroInformatics Association

ACHIEVEMENTS AND INTERNSHIPS

- JEE Advanced 2022 – All India Rank 3424 (Top 0.3 % of 1,60,000 candidates)
- 2025 Summer internship at **National Astronomical Observatory of Japan (NAOJ)** with **Prof. Maria Giovanna Dainotti.**
- Research collaboration with the **Indian Institute of Astrophysics (IIA), Bengaluru since December 2024.**